STUDENT PORTAL EXPERIENCE

School Management System

Technical Specification & Implementation Guide — v1.0 · March 2026

Confidential · Internal Documentation

Audience
Development Team · Product Team · School Administrators

Related Modules
Student Management · Academics · Attendance · Examination · E-Learning · Library · Finance · Communication · Parent Module

---

1. MODULE OVERVIEW

The **Student Portal** is the primary interface through which students interact with the School Management System.

Unlike the Parent Portal, the Student Portal focuses on:

- **daily academic engagement**
- **learning resources**
- **assignment management**
- **school activities**
- **personal academic progress**

The portal is designed to encourage **student autonomy and responsibility** in managing their academic life.

---

2. CORE SUBMODULES

Submodule	Description	Priority	Status
Student Dashboard	Overview of academic life	Core	Required
Academic Management	Subjects, curriculum, assignments	Core	Required
Timetable	Daily class schedule	Core	Required
Attendance Tracking	Student attendance records	Core	Required
Examination & Results	Exam schedules and results	Core	Required
E-Learning	Digital learning resources	Core	Required
Activities & Clubs	Extracurricular participation	Enhanced	Recommended
Library Access	Book search and borrowing	Core	Required
Student Finance (Limited)	Fee status visibility	Core	Required
Communication	Teacher and school messaging	Core	Required

---

3. STUDENT LOGIN & AUTHENTICATION

Students access the system through a secure login.

Login Credentials

Field	Value
Username	Admission Number
Example	STU-2024-001
Password	Temporary password issued during admission

After first login, students may update:

- password
- email
- phone number

Authentication Workflow

```
Login Page
     |
Student Authentication
     |
JWT Token Generated
     |
Redirect to Student Dashboard
```

Session rules:

Rule	Description
Session timeout	30 minutes inactivity
Failed attempts	Lock after 5 attempts
Password reset	Verified via school records

---

4. STUDENT DASHBOARD

The dashboard is the **central hub** of the student experience.

Dashboard Layout

```
Student Dashboard
│
├ Today's Timetable
├ Upcoming Assignments
├ Attendance Summary
├ Recent Grades
├ School Announcements
├ Upcoming Events
└ Learning Resources
```

Widgets should update **in real-time**.

---

5. ACADEMIC MANAGEMENT

The portal provides access to academic information.

Subjects & Courses

Students see all enrolled subjects.

Example:

```
Mathematics
English
Physics
Chemistry
History
```

Each subject page includes:

- syllabus
- assignments
- teacher details
- class resources

---

6. ASSIGNMENT MANAGEMENT

Students manage assignments through the portal.

Assignment Features

Feature	Description
Assignment list	All pending tasks
Due dates	Submission deadlines
File upload	Submit digital assignments
Teacher feedback	Comments and grades

Assignment Workflow

```
Teacher creates assignment
        |
Students receive notification
        |
Student completes work
        |
Upload assignment
        |
Teacher reviews and grades
```

---

7. TIMETABLE MANAGEMENT

The timetable displays the student's daily schedule.

Timetable View

Example daily schedule:

```
08:00 – Mathematics
09:00 – English
10:00 – Physics
11:00 – Break
11:30 – Chemistry
12:30 – History
```

Features include:

- daily view
- weekly view
- classroom location

---

8. ATTENDANCE TRACKING

Students can view their attendance records.

Attendance Information

Data	Description
Attendance percentage	Monthly average
Absences	Recorded absences
Late arrivals	Tardiness records

Example:

```
Attendance Rate
94%

Absences
2 this month
```

Detailed daily attendance logs remain primarily for teachers.

---

9. EXAMINATION & RESULTS

Students access exam information.

Exam Features

Feature	Description
Exam timetable	Upcoming exams
Results	Subject scores
Report cards	Term performance

Example:

```
Mathematics: 85%
English: 79%
Physics: 82%
```

Students may download:

- report cards
- transcripts

---

10. E-LEARNING SYSTEM

The portal integrates digital learning.

Learning Features

Feature	Description
Course materials	PDFs, slides, videos
Recorded lessons	Replay lectures
Online quizzes	Digital assessments
Learning progress	Course completion tracking

Learning content may include videos from:

- YouTube
- Vimeo

---

11. ACTIVITIES & CLUBS

Students may join extracurricular activities.

Activities Module

Examples include:

- sports teams
- debate clubs
- robotics club
- music band

Students can:

```
Browse clubs
Join activities
View event schedules
```

---

12. LIBRARY ACCESS

Students interact with the school library digitally.

Library Features

Feature	Description
Book catalog	Search books
Borrowing status	Track borrowed books
Due dates	Return reminders
Digital resources	eBooks and journals

---

13. STUDENT FINANCE (LIMITED VIEW)

Students have **restricted financial access**.

Visible Information

Students can see:

```
Fee Status
Paid / Pending
```

Students cannot see:

- invoice breakdown
- payment history
- family financial details

Full financial data is accessible only in the **Parent Portal**.

---

14. COMMUNICATION SYSTEM

Students receive communication from the school.

Communication Channels

Students receive:

- announcements
- teacher messages
- assignment alerts
- event notifications

Delivery channels:

```
In-app notification
Email
SMS
```

Messaging services may integrate with:

- Twilio
- Africa's Talking

---

15. SECURITY & ACCESS CONTROL

Student accounts must be protected from unauthorized access.

Security Features

Feature	Description
Role-based access	Students only access their records
Activity logging	Track portal actions
Session monitoring	Detect suspicious access

---

16. ACCESS LOGGING

Every student action is logged.

Logged Events

Event
login_attempt
login_success
assignment_submission
exam_result_view
timetable_view

Log structure:

Field	Description
user_id	Student ID
action	Activity
ip_address	Access device
timestamp	Event time

---

17. API ENDPOINT STRUCTURE

Student portal APIs should use:

```
/api/student/
```

Core APIs

Method	Endpoint	Description
GET	/api/student/dashboard	Dashboard data
GET	/api/student/timetable	Student schedule
GET	/api/student/subjects	Enrolled subjects
GET	/api/student/attendance	Attendance record
GET	/api/student/exams	Exam schedule
GET	/api/student/results	Exam results
POST	/api/student/assignment/upload	Assignment submission

---

18. MODULE INTEGRATIONS

Module	Purpose
Student Management	Student records
Academics	Subject content
Attendance	Attendance data
Examination	Exam scheduling
Finance	Fee status
Library	Book management
Communication	Notifications
Parent Module	Parent visibility

---

19. IMPLEMENTATION WORKFLOW

Daily student usage flow:

```
Student Login
     |
Open Dashboard
     |
Check timetable
     |
Attend classes
     |
Complete assignments
     |
Review grades
     |
Check announcements
```

---

20. DESIGN PRINCIPLE

The Student Portal is designed around:

```
Student Autonomy
Learning Engagement
Academic Accountability
```

The goal is to make the portal **a daily tool students actively use**, not just a place to check grades.